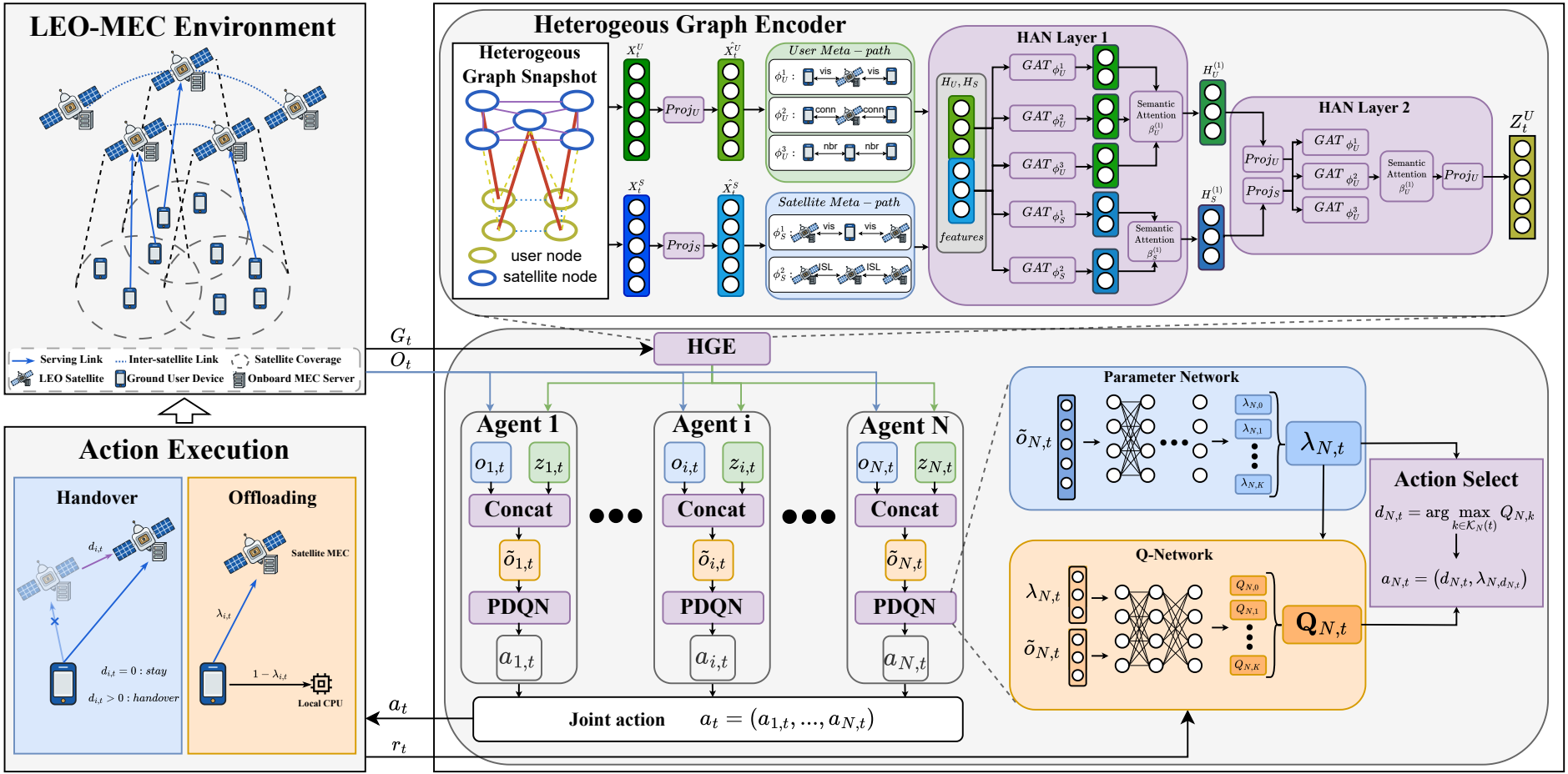

LEO-MEC Environment
Serving Link
Inter-satellite Link
Satellite Coverage
LEO Satellite
Ground User Device
Onboard MEC Server
Action Execution
Handover
$d_{i,t}$
$d_{i,t} = 0 : stay$
$d_{i,t} > 0 : handover$
Offloading
Satellite MEC
$\lambda_{i,t}$
$1 - \lambda_{i,t}$
Local CPU
Heterogeous Graph Encoder
Heterogeous Graph Snapshot
user node
satellite node
X_t^U
$Proj_U$
$\hat{X}_t^U$
User Meta – path
ϕ_U^1 : vis vis
ϕ_U^2 : conn conn
ϕ_U^3 : nbr nbr
X_t^S
$Proj_S$
$\hat{X}_t^S$
Satellite Meta – path
ϕ_S^1 : vis vis
ϕ_S^2 : ISL ISL
HAN Layer 1
H_U, H_S
features
$GAT_{\phi_U^1}$
$GAT_{\phi_U^2}$
$GAT_{\phi_U^3}$
$GAT_{\phi_S^1}$
$GAT_{\phi_S^2}$
Semantic Attention
$\beta_U^{(1)}$
$\beta_S^{(1)}$
$H_U^{(1)}$
$H_S^{(1)}$
HAN Layer 2
$Proj_U$
$Proj_S$
$GAT_{\phi_U^1}$
$GAT_{\phi_U^2}$
$GAT_{\phi_U^3}$
Semantic Attention
$\beta_U^{(1)}$
$Proj_U$
Z_t^U
G_t
O_t
HGE
Agent 1
Agent i
Agent N
$o_{1,t}$ $z_{1,t}$
$o_{i,t}$ $z_{i,t}$
$o_{N,t}$ $z_{N,t}$
Concat
$\tilde{o}_{1,t}$
$\tilde{o}_{i,t}$
$\tilde{o}_{N,t}$
PDQN
$a_{1,t}$
$a_{i,t}$
$a_{N,t}$
Joint action $a_t = (a_{1,t}, ..., a_{N,t})$
a_t
r_t
Parameter Network
$\tilde{o}_{N,t}$
$\lambda_{N,0}$
$\lambda_{N,1}$
$\lambda_{N,K}$
$\lambda_{N,t}$
Q-Network
$\lambda_{N,t}$
$\tilde{o}_{N,t}$
$Q_{N,0}$
$Q_{N,1}$
$Q_{N,K}$
$\mathbf{Q}_{N,t}$
Action Select
$d_{N,t} = \arg\max_{k \in \mathcal{K}_N(t)} Q_{N,k}$
$a_{N,t} = (d_{N,t}, \lambda_{N,d_{N,t}})$